

NISCHAL DEO

+977 9851313347 nischaldeo.com.np nischaldeo@outlook.com [Nischal Deo](https://www.linkedin.com/in/NischalDeo) [nxd010](https://github.com/nxd010)

Education

Motilal Nehru National Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Aug 2023 – May 2027

Prayagraj, Uttar Pradesh

DAV Sushil Kedia Vishwa Bharati

Central Board of Secondary Education +2

April 2020 – May 2022

Kathmandu, Nepal

DAV Sushil Kedia Vishwa Bharati

National Education Board

April 2010 – May 2022

Kathmandu, Nepal

Personal Projects

Shelfies | ReactJS, Flask, Python, YOLOv8, Kaggle

Feb 2025

- Prototyped a commercially-viable, AI-powered inventory system using a YOLOv8 model to automate stock monitoring for over 40+ retail SKUs.
- Built a full-stack dashboard with a Flask API and ReactJS to visualize real-time stock levels, log checkout transactions, and automate updates, significantly reducing manual tracking efforts.

Vayu | ESP32, FastAPI, SQLite and ReactJS

August 2025

- Engineered a full-stack monitoring system using ESP32, FastAPI, and React to track real-time environmental metrics.
- Designed a tiered data retention strategy using SQLAlchemy aggregates, reducing long-term storage requirements while maintaining 14-day historical trends.
- Implemented background tasks for automated database optimization, including index analysis and periodic VACUUMing to ensure query performance under high-frequency ingestion (0.5Hz).

Deal Ninja | Django, BeautifulSoup, Cron, Telegram API

March 2024

- Addressed the challenge of manual price-tracking by designing an automated service for Amazon.com; utilized Django and BeautifulSoup to scrape product data and implemented cron jobs for continuous monitoring, ensuring users never miss a deal.
- Engineered an automated notification system using the Telegram API to instantly alert users of price drops, successfully automating the deal-hunting process.
- Structured a PostgreSQL database using the Django ORM to efficiently store user subscriptions, product details, and historical price data.

Technical Skills and Interests

Languages: Python, C++, JavaScript, C, Bash, PowerShell

Developer Tools: Git, Postman, Docker, Kubernetes

Frameworks: Django, React.js, Flask, Node.js,

Cloud/Databases: PostgreSQL, MySQL, MongoDB

Soft Skills: Communication, Problem-Solving, Leadership, Adaptability

Coursework: Data Structures, Algorithms, AIML, Computer Architecture, Object Oriented Programming

Areas of Interest: Web Development, Android Development, Embedded Systems, System Design

Leadership / Extracurricular

Nepathya Cultural Committee

2024 – Present

Head Coordinator

MNNIT

- Achieved a 2nd runner-up in college fest(Culrav) with over 12 teams
- Managed all the logistical and technical progress.

MNNIT FC

2024 – Present

Coordinator

MNNIT

- Achieved a 1st runner-up in Inter College with over 15+ teams from all over country